

## Project Title

Daily Firewall, Network, and Syslog Automation on Kali Linux



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## **Abstract**

This project automates the daily collection and emailing of firewall logs, network activity, and system logs on Kali Linux using shell scripting and cron jobs. Unlike other solutions, this setup avoids using Python and instead relies on built-in tools (mailx, msmtplib, or ssmtp) for email delivery through Gmail SMTP. The goal is to enhance system visibility and automate monitoring tasks with minimal dependencies.

## **Objectives**

- Automate the collection of critical system logs.

- Schedule log collection daily without manual intervention.
- Send the reports to a specified email securely via SMTP.
- Maintain logs locally and perform automatic cleanup of old reports

## Tools & Technologies Used

| Tools                               | Technologies Used             |
|-------------------------------------|-------------------------------|
| Kali Linux                          | Operating System              |
| Bash Scripting                      | Automating Tasks              |
| Mailx / msmtplib                    | Sending logs via emails       |
| Cron Jobs                           | Job Scheduling                |
| dmesg, ip, netstat, /var/log/syslog | System and network monitoring |

## System Design

The system uses a shell script to automate daily log collection and email reporting. It gathers firewall logs from dmesg, network details via ip a and netstat -tulnp, and system logs from /var/log/syslog. These logs are compiled into a dated report file and emailed using mailx or msmtplib, ensuring administrators receive regular system updates without manual effort.

## Implementation

### Log Report Script

```
#!/bin/bash
```

```
TODAY=$(date +%Y-%m-%d) LOG_DIR="/var/log"
```

```
REPORT_DIR="/var/log/daily_reports"
```

```
REPORT_FILE="$REPORT_DIR/daily_report_$TODAY.txt"
```

```
mkdir -p "$REPORT_DIR"
```

```
echo "==== Firewall Logs (UFW or iptables) =====" > "$REPORT_FILE" dmesg | grep  
-i "ufw\|iptables" >> "$REPORT_FILE"
```

```
echo -e "\n==== Network Status =====" >> "$REPORT_FILE" ip a >> "$REPORT_FILE"  
netstat -tulnp >> "$REPORT_FILE"
```

```
echo -e "\n==== Syslog =====" >> "$REPORT_FILE" cat "$LOG_DIR/syslog" >>  
"$REPORT_FILE"
```

## Cleanup logs older than 7 days

```
find "$REPORT_DIR" -type f -name "*.txt" -mtime +7 -delete
```

## Send email

```
cat "$REPORT_FILE" | mail -s "Daily Kali Logs - $TODAY" your\_email@gmail.com.
```

## Configure Mail Client (mailx + msmtplib)

Install tools:

```
sudo apt install mailutils msmtplib
```

**Edit ~/.msmtplibrc:**

defaults

auth on

tls on

tls\_trust\_file /etc/ssl/certs/ca-certificates.crt

logfile ~/.msmtp.log

account gmail

host smtp.gmail.com

port 587

from [your\\_email@gmail.com](mailto:your_email@gmail.com)

user [your\\_email@gmail.com](mailto:your_email@gmail.com)

password your\_app\_password

account default: Gmail

### **Set permissions:**

chmod 600 ~/.msmtprc

### **Schedule with Cron**

crontab -e

### **Add job:**

\* \* \* /usr/local/bin/daily\_log\_report.sh

### **Testing:**

sudo /usr/local/bin/daily\_log\_report.sh

### **Check:**

- /var/log/daily\_reports/ for logs
- Your email inbox for log summary

## Challenges

- SMTP setup can be tricky with Gmail (use app password)
- Log permissions (script may need sudo)
- Avoiding duplicate or missing logs in daily runs

## Conclusion

This shell-only solution reliably collects and sends system logs with no need for Python or external scripting languages. It is easy to maintain and works well on headless Kali servers or minimal environments.

